

FinTech — UPI Fraud Intelligence

Detecting Organized Crime with Graph Theory & Generative AI

TEAM

Graph Theory Engine

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Track

FinTech & BFSI — TransOrg AgentIQ Datathon

Data Scale: 20,000 Transactions

₹23.92 Crores Processed

Graph Theory Engine

NetworkX algorithms exposing hidden bipartite fraud rings invisible to SQL

Generative AI Copilot

Deterministic LangGraph Text-to-SQL agent targeting the 30-Point Bonus

The Problem: Why Standard Dashboards Fail

The Blind Spot

Basic SQL dashboards only catch isolated incidents. They fail to detect organized, multi-node crime rings.

UPI transactions are bipartite (User → Merchant), hiding direct peer-to-peer money laundering. Standard monitoring tools treat each transaction in isolation, missing the structural patterns that connect them.

The Reality

Fraudsters use shared bank settlement accounts and synthetic PAN clusters to siphon funds. These coordinated techniques are architecturally invisible to row-level SQL queries and conventional BI dashboards.

The Goal

Build an intelligence system that finds the **root architecture** of the fraud, rather than just treating the symptoms. Move from reactive monitoring to proactive infrastructure dismantlement.

Organized fraud rings exploit the structural gap between transaction-level visibility and network-level intelligence — a gap that no SQL dashboard can close.

The Architecture: 4 Pillars of Intelligence

We engineered an enterprise-grade pipeline rather than a simple visualization tool.

Pillar 1

Data Integrity Pipeline — Robust Pandas/Regex cleaning with zero data loss

Pillar 3

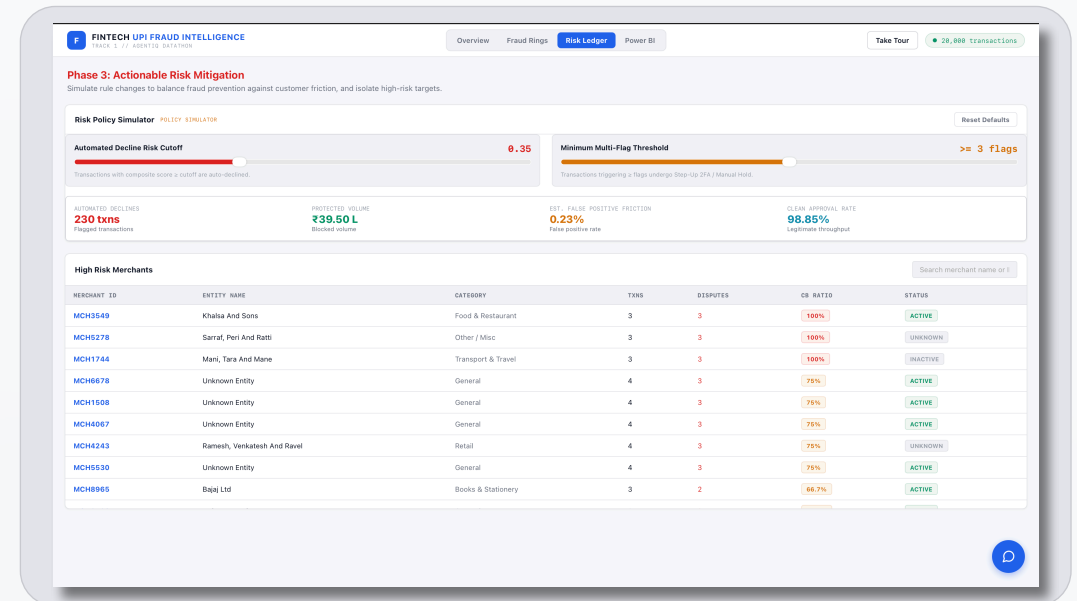
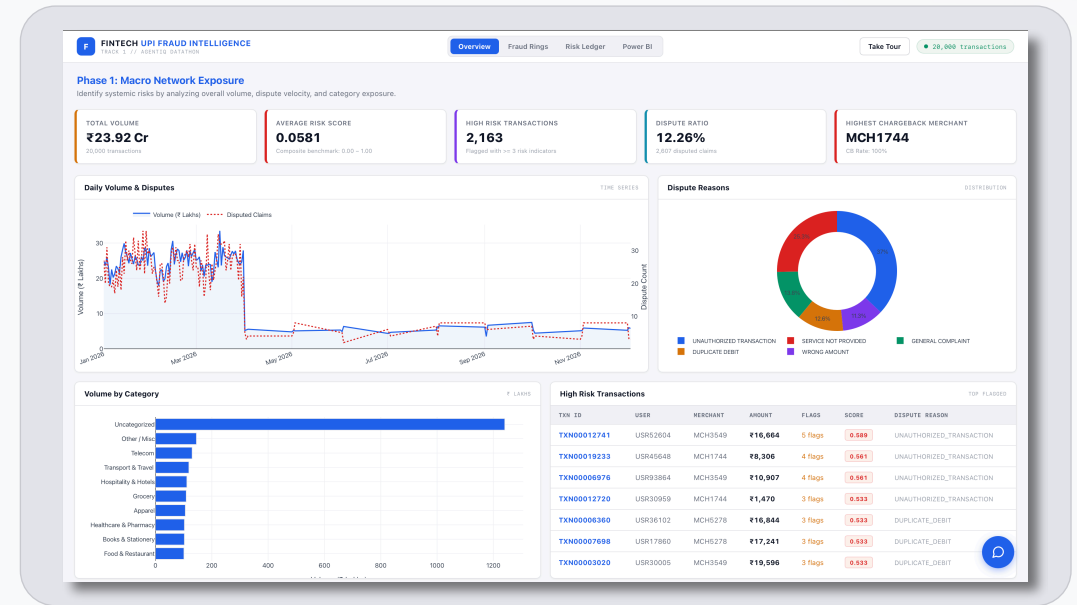
Actionable Risk Simulator — Interactive UI to balance fraud blocking vs. customer friction

Pillar 2

Graph Theory Engine — NetworkX algorithms to find hidden bipartite connections

Pillar 4

Deterministic AI Copilot — Text-to-SQL LangGraph agent (Targeting the 30-Point Bonus)



Pillar 1: Data Integrity & The 5-Way Join

Most pipelines blindly use dropna() on messy datasets. We built a robust pipeline to clean the raw data, fix mismatched IDs, and normalize currencies.

The Challenge

Raw UPI datasets arrive with mismatched transaction IDs, inconsistent currency formats, and schema fragmentation across multiple source tables.

What the pipeline delivers

A clean, traceable pipeline that preserves records, loads analysis-ready data, and creates audit evidence.

Zero Data Loss

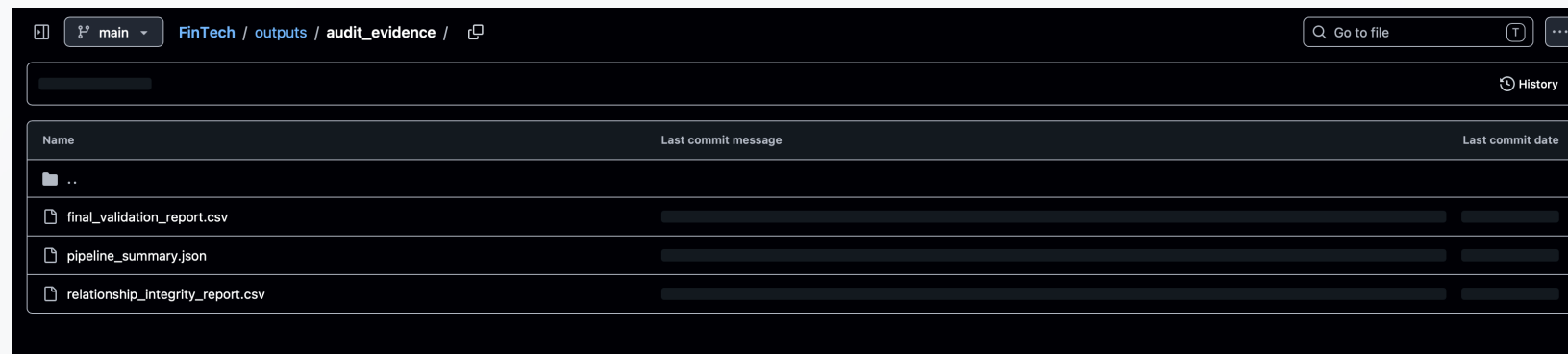
Preserved all 20,000 unique transactions through the 5-way join.

Production Ready

Loaded clean data into in-memory DuckDB for real-time analysis.

Automated Audit Evidence

Generated a cryptographic-style audit log for traceability.



The screenshot shows a file explorer interface with a dark theme. The breadcrumb path is 'FinTech / outputs / audit_evidence /'. Below the path is a table listing files and folders. The table has three columns: 'Name', 'Last commit message', and 'Last commit date'. The files listed are 'final_validation_report.csv', 'pipeline_summary.json', and 'relationship_integrity_report.csv'. There is also a folder icon labeled '..'. A search bar at the top right contains the text 'Go to file'. A 'History' button is located on the right side of the table.

Name	Last commit message	Last commit date
..		
final_validation_report.csv		
pipeline_summary.json		
relationship_integrity_report.csv		

Pillar 2: Graph Theory — The Secret Weapon

Graph theory exposes hidden relationships across merchants, accounts, and disputes that row-level SQL cannot see.

Merchant Collusion

203

Distinct merchants discovered funneling funds into identical bank settlement accounts.

Synthetic Identities

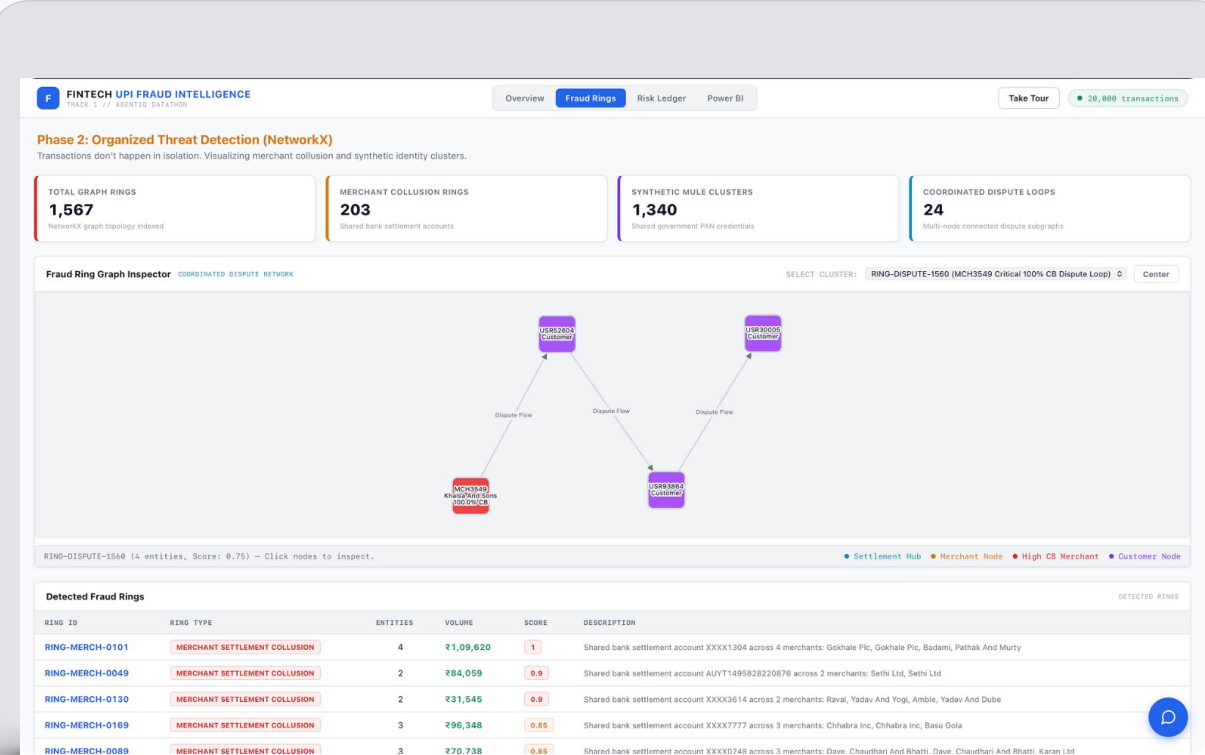
1,340

Clusters of mule accounts detected, registered under the exact same PAN credentials.

Dispute Loops

24

Coordinated high-velocity chargeback loops found.



Graph Theory - Fraud Rings

Pillar 3: Risk Policy Simulator & Power BI

Static charts do not solve business problems. We built interactive tools for executives.

1

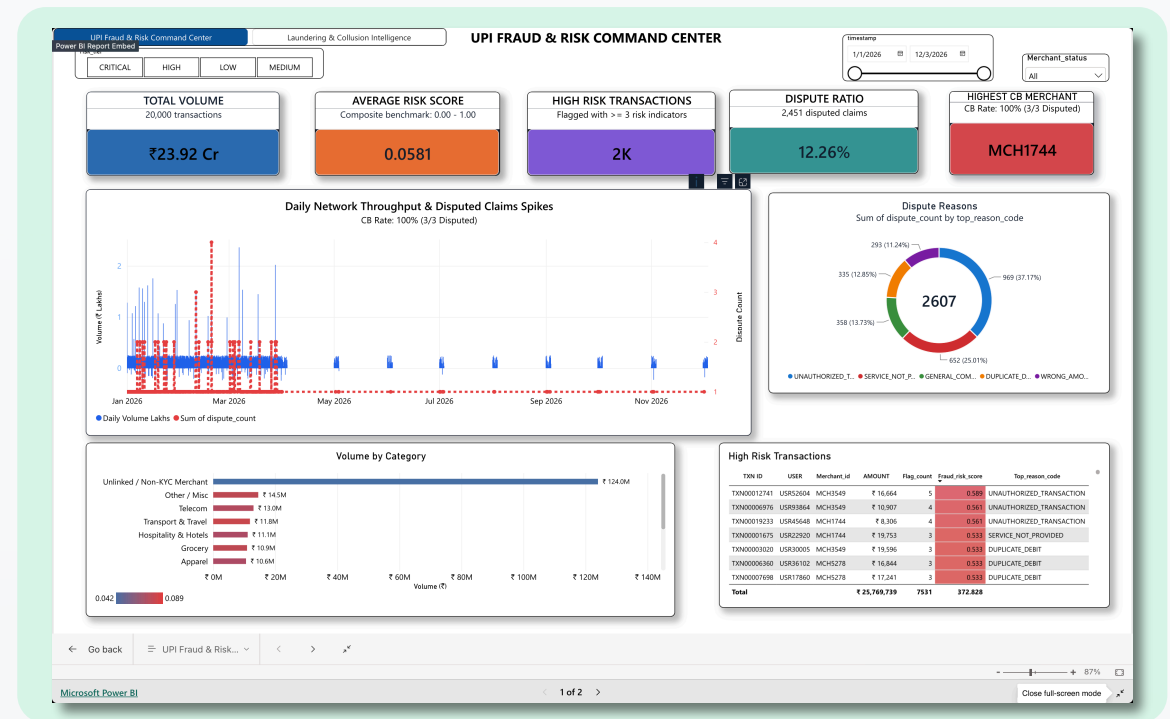
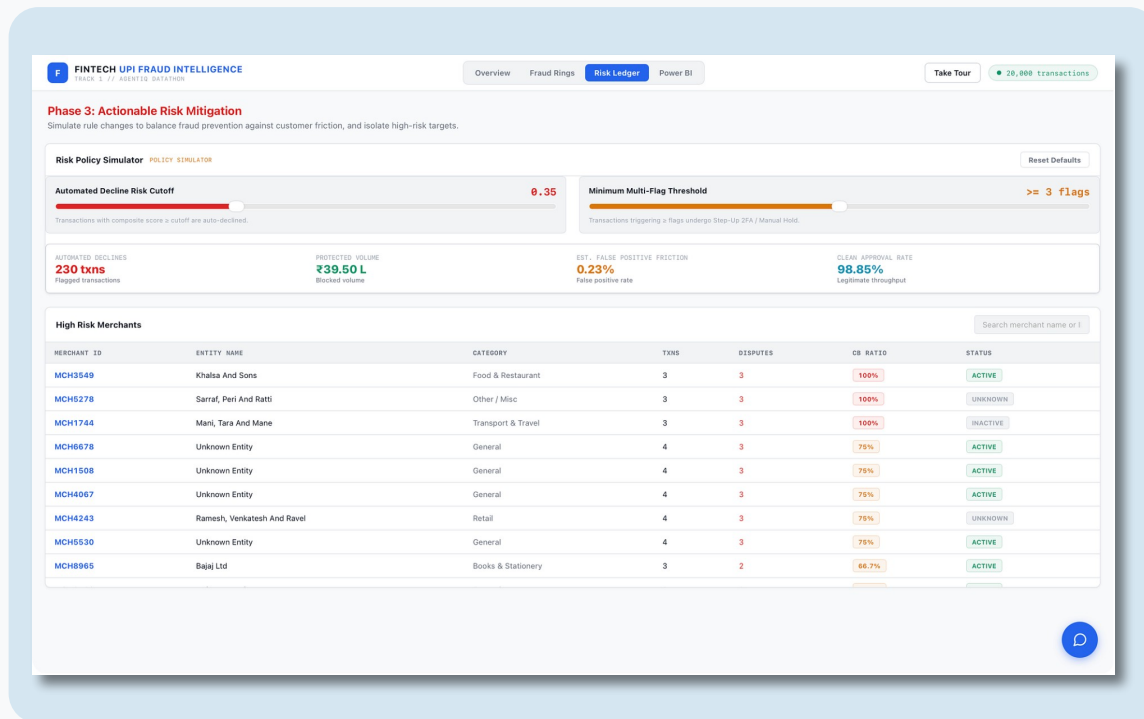
Adjust Thresholds

2

Simulate Trade-off

3

Review Drill-down

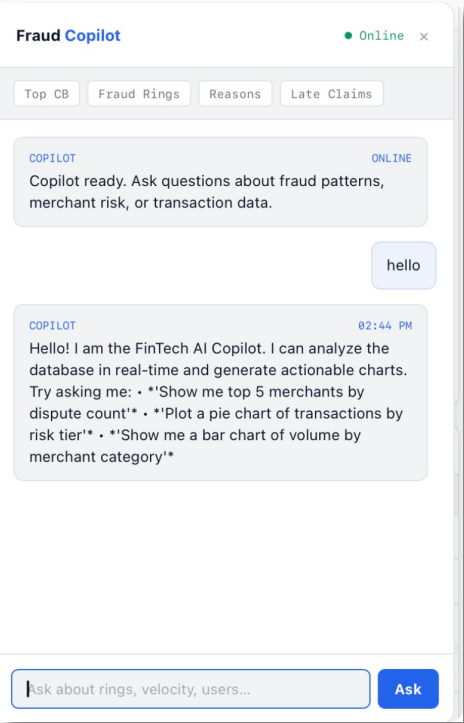
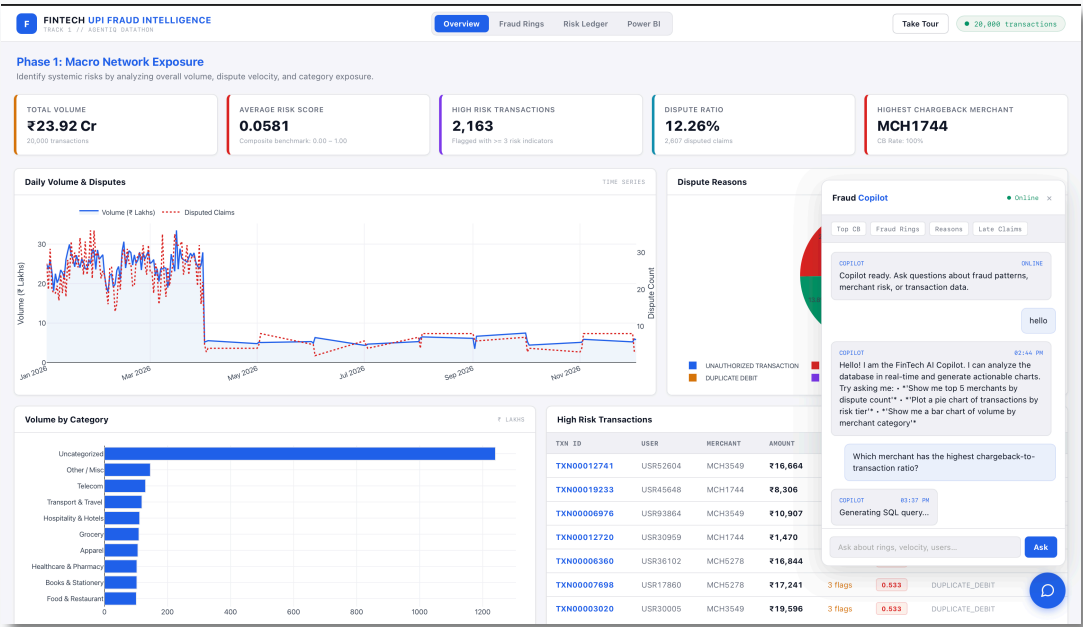


Pillar 4: Deterministic AI Agent — 30-Pt Bonus

We didn't just plug a basic LLM API into a chatbox. We built a **deterministic Text-to-SQL reasoning engine** powered by Groq (Qwen 27B).

AI Agent Highlights

- Plain-English risk questions, no SQL required
- Deterministic DuckDB query generation
- Auditable answers with exact SQL trace



Natural Language

Deterministic SQL

Real-Time Charts

Full SQL Transparency

Enterprise Security & LLM Guardrails

We hardcode strict guardrails so the AI remains safe in a live banking environment.

Anti-SQL Injection

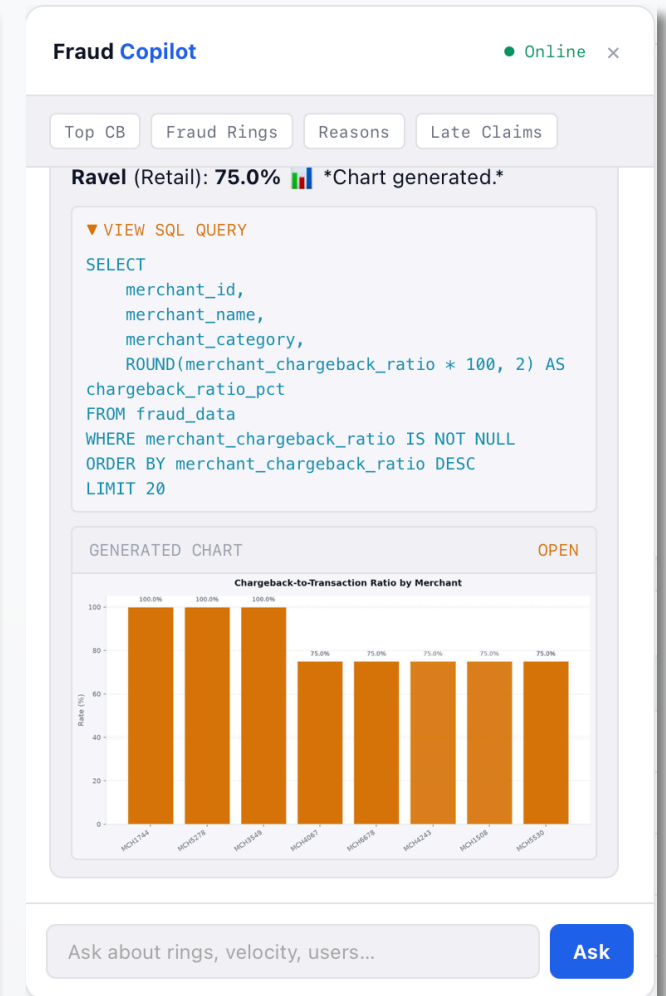
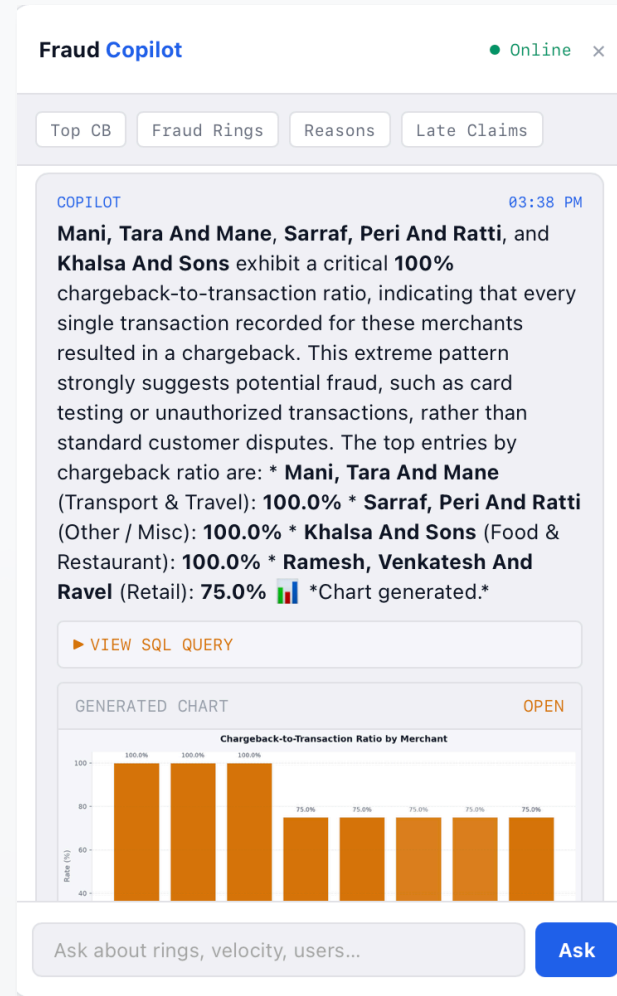
Blocks DML and DDL commands like DROP, DELETE, and UPDATE before execution.

API Resource Protection

Enforces a strict 10-query limit per session to prevent abuse and control infrastructure cost.

Path Traversal Blocks

Validates file extensions to protect backend logic and secrets from unauthorized access.

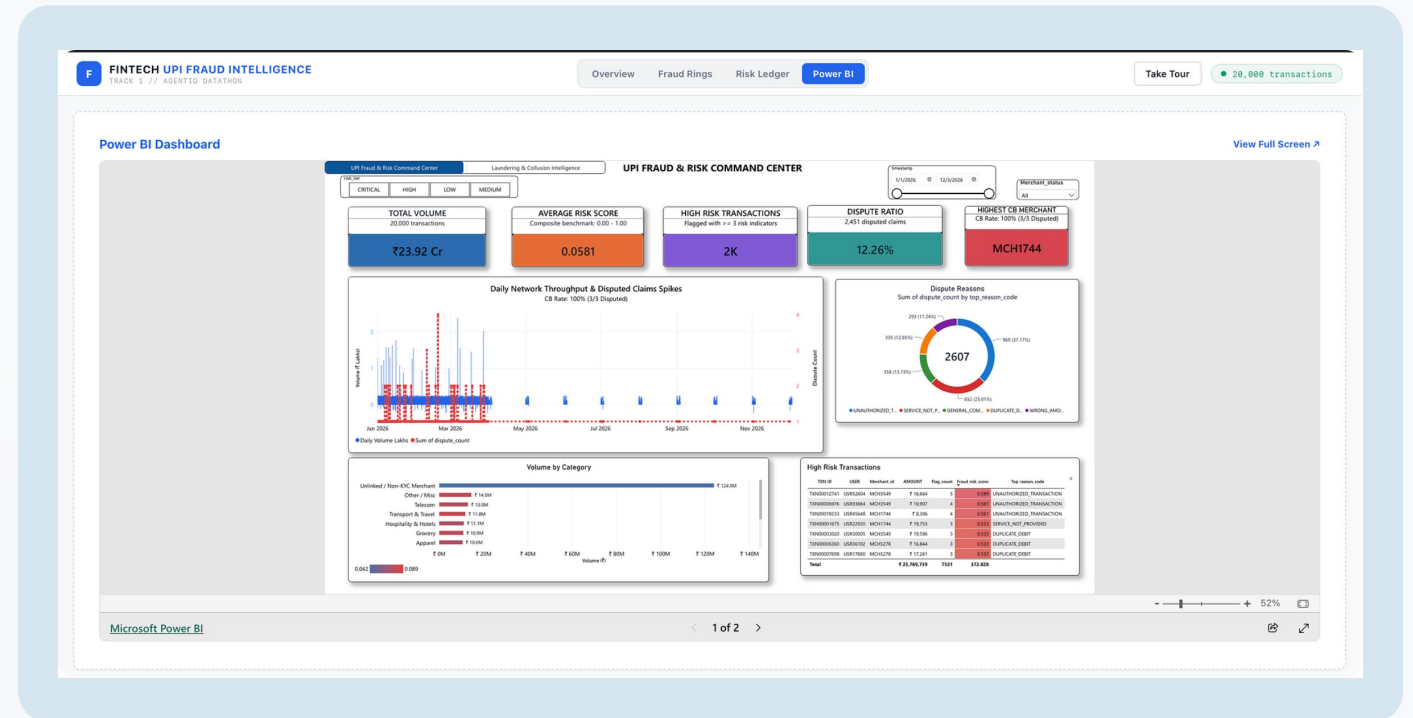


Financial Impact & Results

The platform scanned exactly ₹23.92 Crores in total processed volume.

Platform Outcomes

- Actionable Intelligence — shifted focus from monitoring transactions to dismantling fraud infrastructure.
- Speed — executives query the database in milliseconds with natural language.
- Scalability — in-memory DuckDB and NetworkX scale to millions of rows on edge devices.



Thank You & Live Demo

Live Deployment

<https://fintech-dk9k.onrender.com>

GitHub Repository

<https://github.com/za-aki/FinTech>

The Team

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Graph Theory

- 203 collusion rings
- 1,340 identity clusters
- 24 chargeback loops

AI Copilot

- Deterministic
- LangGraph
- Groq Qwen 27B
- SQL transparency

Security

- Anti-injection
- Guardrails
- Rate limiting
- Path traversal protection